Dong Zhang

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Education

Tianjin University, Tianjin, China 2023 – Present

B.Sc. candidate in Applied Physics, expected June 2027

Research / Project Experience

Study of Electromagnetic Form Factors for Different Nuclear Shapes Mar. 2025 – Mar. 2026

Undergraduate Innovation Project, Project Leader Supervisor: Associate Prof. Baoyi Chen

- Quantified the electromagnetic properties of nuclei with different geometric shapes, aiming to provide a theoretical framework for experimental nuclear-shape reconstruction.
- Studied the Liénard-Wiechert potentials, multipole expansion, and relativistic electromagnetic field theory.
- Combined classical electrodynamics with parametrized nuclear-structure models to go beyond the spherical approximation and predict electromagnetic field distributions for non-standard nuclear geometries.
- Developed simulation programs for electromagnetic field distributions of fast-moving charges and maintained a database of nuclear-shape parameters, including quadrupole and octupole moments, for geometry–field correlation analysis.

Machine Learning Applications in Ill-posed Inverse Problems Jul. 2025 – Aug. 2025

Summer Research Project, The Chinese University of Hong Kong, Shenzhen Supervisor: Associate Prof. Kai Zhou

- Applied machine learning methods to inverse problems in spectral function reconstruction.
- Used supervised and unsupervised learning methods, and developed and maintained machine learning programs in Python.
- Achieved good fitting and prediction performance in single-peak spectral reconstruction cases.

Study of Excited Bottom and Charmed Hadrons May 2025 – May 2026

Research Training Program, University of Chinese Academy of Sciences Supervisor: Prof. Jibo He

- Participated in high-energy physics data analysis related to heavy-hadron excited states.
- Learned the basic structure of large particle colliders and acquired basic Linux operation skills.
- Used ROOT and C++ for data analysis tasks.

Computations of Cosmological Correlation Functions Apr. 2026 – Prensent

Research Project, China University of Petroleum Supervisor: Jiaqi Chen

- Studied tree-level cosmological correlation functions in de Sitter spacetime, with emphasis on Feynman rules, propagator structures, and boundary conditions in the Schwinger-Keldysh, or in-in, formalism.
- Investigated a computational framework for cosmological correlators with multiple massive exchange fields based on IBP reduction and differential equations in d-log form.
- Built a numerical solving and validation workflow in Mathematica for three-vertex tree diagrams.

Technical Skills

- Programming Languages: Python, C++
- Research Tools: Mathematica, ROOT
- Platform: Linux

Additional Information

- Language: Proficient in English, CET-6: 506